Cartesian Fibrations

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Last time we discussed how we can model presheaves into spaces, by using right fibrations. This time we will discuss how we can model presheaves into higher categories, by discussing Cartesian fibration. We first give an example to motivate this discussion. Then we introduce two ways to think about Cartesian fibrations. One can be found in [dB16]. The other will follow work by Mazel Gee in [Mg15], but translated to the context of complete Segal spaces. For more information and a thorough discussion of Cartesian fibrations see Section 2.4 of [Lu09] in the context of quasi-categories. At the end we will see some examples, which justify thinking of Cartesian fibrations as maps into higher categories.

Motivating Example

Let us start by discussing an example from the realm of categories. One thing that geometers are interested in is the theory of vector bundles over geometric objects. This motivates following definitions.

Definition 1.1. Let $\mathcal{M}an$ be the category of manifolds. Objects are manifolds and maps are smooth maps of manifolds.

Definition 1.2. There is a map of objects $Vect : \mathcal{M}an \to \mathcal{C}\mathrm{at}$ which takes each object $\mathcal{M}$ to the category of vector bundles over $\mathcal{M}$, denoted by $Vect(\mathcal{M})$.

Why did we not say the map is a functor? Turns out the map wants to be a functor, but something very subtle goes wrong.

Remark 1.3. For each map of manifolds $f : \mathcal{M} \to \mathcal{N}$, we can define a pullback map

$$f^* : Vect(\mathcal{N}) \to Vect(\mathcal{M})$$

which takes each vector bundle $\mathcal{V} \to \mathcal{N}$ to the pullback vector bundle $f^*\mathcal{V} \to \mathcal{M}$. The existence of this construction suggests functoriality except for the following fact:

For each two maps of manifolds $f : \mathcal{M} \to \mathcal{N}$ and $g : \mathcal{N} \to \mathcal{O}$, we get maps

$$g^* : Vect(\mathcal{O}) \to Vect(\mathcal{N}) \text{ and } f^* : Vect(\mathcal{N}) \to Vect(\mathcal{M})$$

The theory of vector bundles tells us that there is an equivalence

$$(g \circ f)^* \cong f^* \circ g^*$$

but the map is not the identity.

So, functoriality only holds *up to isomorphism*. This suggests two possible solutions:

1. Consider maps only *up to isomorphism*. This will work but some information will be lost.
2. Find a way to keep track of all the isomorphism information as part of the maps.

The second approach leads us to think about fibrations instead of functors. This suggests following definition

Definition 1.4. Let $\mathcal{B}un$ be the category of vector bundles over a manifolds. Concretely, objects are maps $\mathcal{V} \to \mathcal{M}$ where $\mathcal{M}$ is a manifold and $\mathcal{V}$ is a vector bundle over $\mathcal{M}$. A map is a commutative diagram

$$\begin{tikzcd} W \arrow[r] \arrow[d] & V \arrow[d] \\ M \arrow[r] & N \end{tikzcd}$$

There is a natural forgetful functor

$$U : \mathcal{B}un \to \mathcal{M}an$$

which takes each object $\mathcal{V} \to \mathcal{M}$ to $\mathcal{V}$.

Remark 1.5. Note that over each object $\mathcal{M}$ there is the category of vector bundles over the manifold $\mathcal{M}$, which is exactly the image of the map $Vect(\mathcal{M})$. So, the map $\mathcal{B}un$ preserves all the information of $Vect$ but also has some additional information to keep track of the generalized functoriality.

The functor U is what we call a *functor fibered in categories*. Over each point the fiber is a category and it satisfies suitable compatibility conditions which guarantee us that we have functoriality up to isomorphisms.

Our goal for this talk is to discuss this idea in the context of higher categories.

Reminder on Right Fibrations

Recall from [We17] that a map $f : Y \to X$ of simplicial spaces is a right fibration if the following is a homotopy pullback diagram

$$\begin{tikzcd} R^{F(1)} \arrow[r, "f^{F(1)}"] \arrow[d, "t"'] \arrow[dr, phantom, "\ulcorner"] & X^{F(1)} \arrow[d, "t"] \\ R \arrow[r, "f"'] & X \end{tikzcd}$$

where the map $t : X^{F(1)} \to X$ is the map that takes each arrow to its target. Concretely this means that the map:

$$R^{F(1)} \to X^{F(1)} \times_X R$$

is a Reedy equivalence.

Intuitively it suggests that R does not carry any additional information beyond the object level as every higher level is already determined by the objects of R and the higher morphisms of X (for more details see Section 4 of [We17]). Thus this definition is not a suitable generalization of a functor fibered in categories, as any reasonable generalization should allow us to have non-trivial maps in the fiber.

Simplicial Functors and Complete Segal Objects

Before we can discuss what kind of fibrations would be able to model maps into higher categories, it is quite insightful to look into the case of higher category valued functors.

Definition 3.1. Let $\mathcal{C}$ be a simplicial category. One basic model of a higher category valued functor would be a simplicial space valued functor

$$F : \mathcal{C}^{op} \to s\mathcal{S}$$

such that for each object $x \in \mathcal{C}$ the image $F(c)$ is actually a complete Segal space.

Let us think about that definition. The functor F maps into simplicial spaces. That means there are space valued functors $F_0 : \mathcal{C}^{op} \to \mathcal{S}$, $F_1 : \mathcal{C}^{op} \to \mathcal{S}$, $F_2 : \mathcal{C}^{op} \to \mathcal{S}$, ... such that those functors form a simplicial object in the category $Fun(\mathcal{C}^{op}, \mathcal{S})$. But there is more, the image at each point is a complete Segal spaces. Thus the the simplicial object must satisfy some relations. The good thing is the relations are just some limit conditions which exist in the category of presheaves on $\mathcal{C}$. Concretely, we want the simplicial object in functors to satisfy following two conditions:

Segal Condition: The map

$$F_n \xrightarrow{\simeq} F_1 \times_{F_0} ... \times_{F_0} F_1$$

is an equivalence of simplicial functors. In other words for each object $c \in \mathcal{C}$, the map

$$F_n(c) \xrightarrow{\simeq} F_1(c) \times_{F_0(c)} \dots \times_{F_0(c)} F_1(c)$$

is a Kan equivalence

Completeness Condition: The map

$$F_0 \xrightarrow{\simeq} F_1 \times_{F_1 \times_{F_0} F_1 \times_{F_0} F_1} F_3$$

is an equivalence of simplicial functors. In other words for each object $c \in \mathcal{C}$, the map

$$F_0(c) \xrightarrow{\simeq} F_1(c) \times_{F_1(c) \times_{F_0(c)} F_1(c) \times_{F_0(c)} F_1(c)} F_3(c)$$

There is a word for that, namely a *complete Segal objects*. More precisely we have following definition.

Definition 3.2. Let $\mathcal{C}$ any notion of a higher category with finite limits. A complete Segal object is a simplicial object $X_\bullet \in Fun(\Delta^{op}, \mathcal{C})$ such that it satisfies the *Segal condition* and the *completeness condition*.

Remark 3.3. We have not discussed limits yet, but it will not matter for the cases we are considering as the notion of limit will be clear in those cases.

Thus, we can say the following sentence:

A functor into higher categories is a complete Segal object in functors into spaces

This realization will give us our first way to think about Cartesian fibrations.

Definition of Cartesian Fibrations as Complete Segal Object in Right Fibratios

We will now give our first definition of a Cartesian fibrations. As we discussed we can model space valued presheaves by right fibrations. It has been discussed to some extent in Section 3 of [dB16]. Thus the following should be true:

Vague Definition 4.1. A *Cartesian fibration* over X is a complete Segal object in right fibrations over X.

Let us be a little bit more precise.

Definition 4.2. Let X be a simplicial space. The category $s\mathcal{S}_{/X}$ has as objects maps $Y \to X$ and morphisms commuting triangles of such maps.

Definition 4.3. A simplicial object in $s\mathcal{S}_{/X}$ is a simplicial object in $s\mathcal{S}$ such that all maps commute with composition down to X. It can be depicted as follows:

$$\begin{tikzcd}
R_0 \arrow[r, shift left, leftarrow, "d_1"] \arrow[r, shift right, leftarrow, "d_0"'] \arrow[dr, bend right, "p_0"'] & R_1 \arrow[r, shift left, leftarrow, "d_1"] \arrow[r, shift right, leftarrow, "d_0"'] \arrow[dr, "p_1"] & R_2 \arrow[r, shift left, leftarrow] \arrow[r, shift right, leftarrow] \arrow[d, "p_2"] & R_3 \arrow[r, shift left, leftarrow] \arrow[r, shift right, leftarrow] \arrow[dl, "p_3"] & \cdots \arrow[dll, bend left, "p_4"] \\
& & X & &
\end{tikzcd}$$

Now we can give following definition

Definition 4.4. The map above is a *Cartesian fibration* if for every i the map $R_i \to X$ is a right fibration and if it satisfies following two conditions

Segal Conditions:

$$R_n \to R_1 \times_{R_0} ... \times_{R_0} R_1$$

is a Reedy equivalence.

Completeness Condition:

$$R_0 \to R_3 \times_{R \times_{R_0} R_1 \times_{R_0} R_1} R_1$$

Let us try to set up a categorical framework for that.

Definition 4.5. Let $ss\mathcal{S}$ be the category of *bisimplicial spaces*. Thus each object is a grid of spaces Y_{kn}, where $k \geq 0$ and $n \geq 0$. A diagram of such simplicial space can depicted as:

$$\begin{array}{cccc}
Y_{00} & Y_{10} & Y_{20} & ... \\
Y_{01} & Y_{11} & Y_{21} & ... \\
Y_{02} & Y_{12} & Y_{22} & ... \\
\vdots & \vdots & \vdots &
\end{array}$$

Definition 4.6. Let X be a simplicial space. We can think of a simplicial space as a bisimplicial space by defining it $X_{kn} = X_n$. Thus is looks like the following:

$$
\begin{array}{cccc}
X_0 & X_0 & X_0 & \dots \\
X_1 & X_1 & X_1 & \dots \\
X_2 & X_2 & X_2 & \dots \\
\vdots & \vdots & \vdots &
\end{array}
$$

We can now define Cartesian fibrations as follows

Definition 4.7. (*Bisimplicial Space Definition)* A *Cartesian fibration* is a complete Segal object in the category $ss\mathcal{S}_{/X}$, denoted as $C \to X$, such that for each i, $C_i \to X$ is a right fibration.

So, in some sense a Cartesian fibration is to a right fibration, what a complete Segal space is to a space. The idea is that we should be able to deduce many properties of a Cartesian fibrations by studying this comparison. First, we start with an example to get a first feeling.

Example 4.8. Let $X = F(0)$. Let $C \to X$ be a Cartesian fibration. Then the Cartesian fibration condition on $ss\mathcal{S}_{/F(0)}$ implies that for each $k \in \mathbb{N}$, $C_{k\bullet}$ is a space, as it is a right fibration over the point (Example 4.5 in [We17]) Thus, all we need to look focus on is the first row $C_{\bullet 0}$. Now the complete Segal condition implies that this simplicial space is complete Segal space. Thus a Cartesian fibration over the point is just a complete Segal space, which is the same as a map from the point to the category of complete Segal spaces.

This example also implies that in a Cartesian fibration the fiber over each point is a complete Segal space, which is exactly what we would expect.

Now let us try to expand our analogy. We have following fact about complete Segal spaces.

Lemma 4.9. *A complete Segal space W is a space (homotopically constant) if and only if the degeneracy map*

$$s_0 : W_0 \to W_1$$

is a weak Kan equivalence.

This generalizes nicely to the context of Cartesian fibrations

Theorem 4.10. *A Cartesian fibration $C \to X$ is a right fibrations if and only if the map*

$$s_0 : C_0 \to C_1$$

is a Reedy equivalence.

Remark 4.11. Note this condition can be relaxed but we will not focus on that.

Definition of Cartesian Fibrations between Complete Segal Spaces

In this section we will discuss how we can define Cartesian fibrations between complete Segal spaces. Essentially we discuss Definition 2.1 of [Mg15], which is motivated by Section 2.4 in [Lu09], but we will do so in the context of complete Segal spaces.

Definition 5.1. Let $p : W \to V$ be a map of complete Segal spaces. A map $f : x \to y$ in W called is p-Cartesian (or simply cartesian) if the following is square is a homotopy pullback square

$$\begin{tikzcd} W_{/x} \arrow[r, "f_!"] \arrow[d, "p"'] \arrow[dr, phantom, "\ulcorner"] & W_{/y} \arrow[d, "p"] \\ V_{/p(x)} \arrow[r, "p(f)_!"'] & V_{/p(y)} \end{tikzcd}$$

If this condition is satisfied, we say f is a p-Cartesian lift for $p(f)$.

Definition 5.2. (*Simplicial Space Definition*) Let $p : W \to V$ be a Reedy fibration between complete Segal spaces. We say p is a Cartesian fibration, if for each map $g : a \to b$ in V and each object $y \in W$ such that $p(y) = b$ there exists a p-Cartesian lift of g which has codomain y. More concisely we can say, in the map

$$W_1 \to W_0 \times_{V_0} V_1$$

each object on the right hand side can be lifted to a p-Cartesian map in W_1.

Example 5.3. Remember that if $p : W \to V$ is a right fibration, then the map

$$W_1 \xrightarrow{\simeq} W_0 \times_{V_0} V_1$$

is just a weak equivalence. It is not difficult to show that the essentially unique lift that exists is p-Cartesian. By the uniqueness, we can actually deduce that every lift in a right fibration is p-Cartesian.

Remark 5.4. In fact the opposite of the statement above is also true. Thus, a Cartesian fibration $W \to V$ is a a right fibation if and only if every lift is p-Cartesian.

Comparing the Two Definitions

Let us now try to reconcile the two definitions for Cartesian fibrations we have given in this and the last section.

The key question is the following: How do we reconcile the fact that the first definition is defined for a bisimplicial space, whereas the second one is only

defined for simplicial spaces? It seems that we have far more information in a bisimplicial space. The reality is, a lot of data in the first definition is actually redundant. Before we start, recall following construction from [Re17].

Definition 6.1. Let S be a simplicial set. Then we can construct a simplicial space $\Gamma(S)$, which is defined as follows

$$\Gamma(S)_n = (S^{\Delta[n]})^{core}$$

The idea of the construction is that each level is the space of equivalences of n-maps (where we think of objects as 0-maps). We had following result about this construction:

Theorem 6.2. *If Q is a quasi-category then $\Gamma(Q)$ is a complete Segal space.*

The way to think about the simplicial and bisimplicial version of a Cartesian fibrations should be thought of as the relation between a quasi-category and a complete Segal space, where the additional axis keeps track of equivalences of that level. Concretely, we have the following construction:

Definition 6.3. Let $p : C \to X$ be a Cartesian fibration. There exists a maximal right fibration $q : RFib(C) \to X$ with a map $i : RFib(X) \hookrightarrow C$ such that following diagram commutes

$$\begin{tikzcd} RFib(X) \arrow[rr, hook, "i"] \arrow[dr, "q"'] & & C \arrow[dl, "p"] \\ & X & \end{tikzcd}$$

and $RFib(X)$ is a maximal with respect to this property. Concretely $RFib(X)$ is just the simplicial space generated by all p-Cartesian arrows inside C.

Definition 6.4. Let $C \to X$ be a Cartesian fibration between complete Segal spaces. We define the simplicial space $sCart(C)$ over X at each level in the following way:

$$tCart(C)_k = RFib(C^{F(k)})$$

So, the bisimplicial space will look like the following:

$$\begin{array}{cccc} RFib(C)_0 & RFib(C^{F(1)})_0 & RFib(C^{F(2)})_0 & \dots \\ RFib(C)_1 & RFib(C^{F(1)})_1 & RFib(C^{F(2)})_1 & \dots \\ RFib(C)_2 & RFib(C^{F(1)})_2 & RFib(C^{F(2)})_2 & \dots \\ \vdots & \vdots & \vdots & \end{array}$$

It is very easy to see that for each k the map $RFib(C^{F(k)}) \to X^{F(k)}$ is a right fibration. Moreover, using the complete Segal condition on C, we can show that $sCart(C)$ satisfies the complete Segal condition as well. This leads us to the following definition:

Definition 6.5. Let $C \to X$ be a Cartesian fibration between complete Segal spaces. Let $sCart(X)$ be the following pullback:

$$\begin{tikzcd} sCart(X) \arrow[r] \arrow[d, "p"'] \arrow[dr, phantom, "\ulcorner"] & tCart(X) \arrow[d, "q"] \\ X \arrow[r] & X^{F(\bullet)} \end{tikzcd}$$

Then the map $p : sCart(X) \to X$ is a Cartesian fibration of bisimplicial spaces.

Thus we have explicitly found a way to construct a bisimplicial Cartesian fibration from a simplicial Cartesian fibration, generalizing the relation between quasi-categories and complete Segal spaces.

Remark 6.6. An interesting question is whether we can do the opposite construction. In other words take a bisimplicial Cartesian fibration and use some sort of diagonal argument to construct a simplicial Cartesian fibration. This is left as an exercise for the interested reader.

Examples of Cartesian Fibrations

Having seen these definitions let us see couple important examples.

Example 7.1. First let us generalize the example we discussed before. We use the bisimplicial model here as it makes things clear. Let $C \to F(1)$ be a bisimplicial Cartesian fibration. This gives us a collection of right fibrations

$$R_k \to F(1)$$

From the previous talk ([We17]) we know that a right fibration $R_k \to F(1)$ is just the data of a map of spaces $R_k^1 \to R_k^0$. We have these maps for each k. So, we can form a map of simplicial spaces

$$R_\bullet^1 \to R_\bullet^0$$

such that each one of these simplicial spaces is a complete Segal space (this follows from the complete Segal condition that we imposed on our Cartesian fibrations). Thus, it is just a map of complete Segal spaces, which is exactly what a functor from the line into higher categories should be.

Let's look at a more interesting example which is arguably among the most important Cartesian fibrations.

Example 7.2. Let W be a complete Segal space with pullback (again we haven't yet discussed what that means, but we will use a vague definition which works for our purposes). Then $W^{F(1)}$ is again a complete Segal space and there is a target map $t : W^{F(1)} \to W$, which takes each arrow to it's target. The Cartesian condition now reduces to the following. For a map $x \to y$ in W and a map $z \to y$ over y, the following diagram

$$\begin{tikzcd} & z \arrow[d] \\ x \arrow[r] & y \end{tikzcd}$$

can be extended to a square such that the square is t-Cartesian. Given the conditions there is an obvious extension, namely the pullback. The universality of the pullback will then imply that the square is t-Cartesian.

The question we might ask ourselves is: What higher category valued functor is this Cartesian fibration modeling? In order to see that we simply look at the fiber over each point. Let $x \in X$ be an object. Pulling back by that object we get:

$$\begin{tikzcd} X_{/x} \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner"] & X^{F(1)} \arrow[d, "t"] \\ F(0) \arrow[r, "x"'] & X \end{tikzcd}$$

So, over each point x, the fiber is just the over complete Segal space $X_{/x}$ of maps into x. Thus it is modeling the functor that maps each point to $X_{/x}$ and each map $f : x \to y$ to the pullback induced map:

$$f^* : X_{/y} \to X_{/x}$$

Here is also where we need the existence of a pullback in the context of functors.

A Word on coCartesian Fibrations

Similar to the case of left and right fibrations, there is a covariant version of Cartesian fibrations, called coCartesian fibrations. We will here state the two definitions of a coCartesian fibration, analogous to the case of Cartesian fibrations. All the results then can be translated to the language of coCartesian fibrations.

First, the bisimplicial definition.

Definition 8.1. A map $C \to X$ from a bisimplicial space into X is a *coCartesian fibration* if for every i the map $C_i \to X$ is a left fibration and if it satisfies following two conditions

Segal Conditions:

$$C_n \to C_1 \times_{C_0} ... \times_{C_0} C_1$$

is a Reedy equivalence.

Completeness Condition:

$$C_0 \to C_3 \times_{C \times_{C_0} C_1 \times_{C_0} C_1} C_1$$

Now the simplicial definition

Definition 8.2. Let $p : W \to V$ be a map of complete Segal spaces. A map $f : x \to y$ in W called is p-coCartesian (or simply coCartesian) if the following is square is a homotopy pullback square

$$\begin{tikzcd} W_{y/} \arrow[r, "f^*"] \arrow[d, "p"'] \arrow[dr, phantom, "\ulcorner"] & W_{x/} \arrow[d, "p"] \\ V_{p(y)/} \arrow[r, "p(f)^*"'] & V_{p(x)/} \end{tikzcd}$$

If this condition is satisfied, we say f is a p-Cartesian lift for $p(f)$.

Definition 8.3. (*Simplicial Space Definition)* Let $p : W \to V$ be a Reedy fibration between complete Segal spaces. We say p is a coCartesian fibration, if for each map $g : a \to b$ in V and each object $x \in W$ such that $p(x) = a$ there exists a p-coCartesian lift of g which has domain x. More concisely we can say, in the map

$$W_1 \to W_0 \times_{V_0} V_1$$

each object on the right hand side can be lifted to a p-coCartesian map in W_1.

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